Efficient PAC Learning of Distributions under Finite Symmetries: Algorithms for Folded Normals and Gaussian Mixtures

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Abstract

We develop a finite-sample (PAC-style) learning framework for parametric distribution families that are non-identifiable due to a finite symmetry group action. Building on the quotient-space viewpoint and orbit metric $d_{\mathcal{G}}$ used for asymptotic orbit-consistency in [1], we define *PAC orbit-learning* and give sample-complexity bounds controlled by the metric entropy of the quotient $\Theta/\mathcal{G}$. We propose a *discretized sieve ERM* algorithm that performs empirical risk minimization over a finite γ-net of a compact sieve in $\Theta/\mathcal{G}$, yielding an efficient orbit-PAC learner under envelope/Lipschitz conditions. We specialize the results to the folded normal model ($\mathcal{G} = \{\pm 1\}$) and finite Gaussian mixtures ($\mathcal{G} = \mathfrak{S}_k$), and discuss polynomial-time improper and spectral relaxations when global likelihood maximization is computationally intractable [10, 7, 8, 9].

1 Experiments

To validate the theoretical sample complexity bounds, we evaluate the estimators for folded normal distributions and Gaussian mixtures on synthetic data. The experiments are implemented in both Python and R to serve as cross-reference implementations.

1.1 Folded Normal

For the folded normal model, we compare the Net-ERM estimator against the classical moment-based estimator. Data is generated by sampling $X_i \sim \mathcal{N}(\mu, \sigma^2)$ and setting $Y_i = |X_i|$ with parameters $\mu = 1.0, \sigma = 1.0$. For the Net-ERM, we construct a discrete grid of candidates on the quotient space (since

the sign of μ is unidentifiable) and select the minimizer of the negative log-likelihood. The moment estimator solves for μ^2 and σ^2 using the empirical second and fourth moments.

Figure 1 plots the orbit error metric $d_{\mathcal{G}}(\hat{\theta}, \theta^\star)$ versus the sample size n. The error scales as $\mathcal{O}(n^{-1/2})$, confirming the ε^{-2} scaling in the sample complexity bounds.

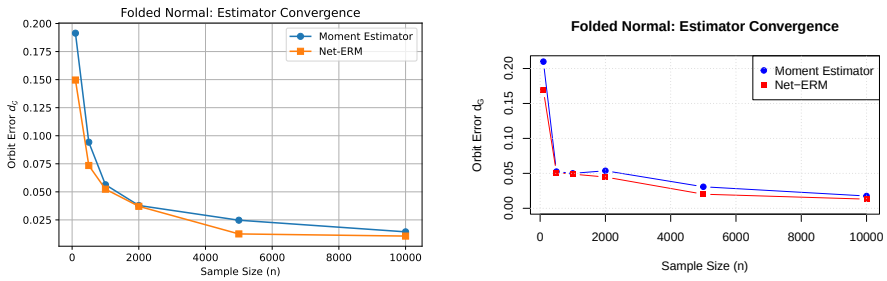

Figure 1: Orbit error $d_{\mathcal{G}}$ vs. Sample Size n for the Folded Normal model. Left: Python implementation. Right: R implementation.

1.2 Gaussian Mixtures

For the Gaussian mixture model, we consider a $k = 2$ component mixture in 1D with true parameters $w = [0.4, 0.6]$, $\mu = [-2.0, 2.0]$, and $\sigma = [1.0, 1.0]$. The components are identifiable only up to permutation in $\mathfrak{S}_2$.

We evaluate the standard Expectation-Maximization (EM) algorithm as a practical, improper relaxation of the exact Net-ERM, computing the permutation-invariant orbit distance $d_{\mathcal{G}}(\hat{\theta}, \theta^\star)$. The convergence of the orbit error as n increases is shown in Figure **??** (Python) and Figure **??** (R).

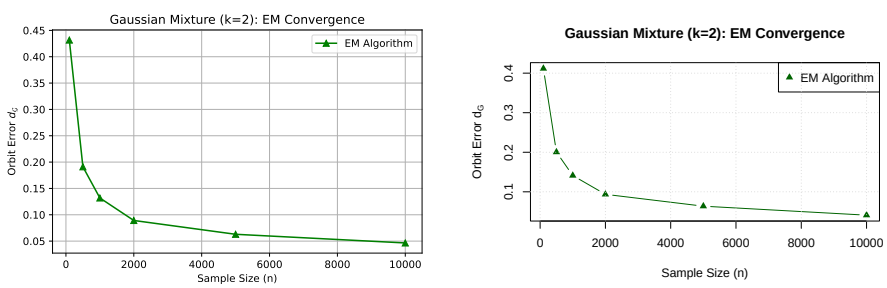

Figure 2: Permutation-invariant orbit error $d_{\mathcal{G}}$ vs. Sample Size n for the $k = 2$ Gaussian Mixture model. Left: Python implementation. Right: R implementation.

2 Discussion

We conclude with open problems on sharp minimax rates on quotient spaces and computational-statistical gaps.

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